

2 More farmers are using environmentally friendly fertilisers.

One example is vermicompost leachate (VCL), a liquid formed when worms decompose organic matter.

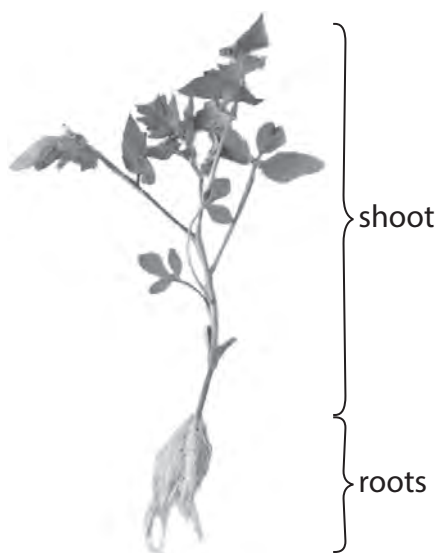
Hoagland's solution contains the mineral ions required for plant growth.

Three mineral-deficient solutions were made from Hoagland's solution:

- solution A had no nitrate ions (no N)
- solution B had no phosphate ions (no P)
- solution C had no potassium ions (no K).

The effect on the dry mass and length of shoots of tomato plants treated with these mineral-deficient solutions, supplemented with VCL, was investigated.

The photograph shows a tomato plant.

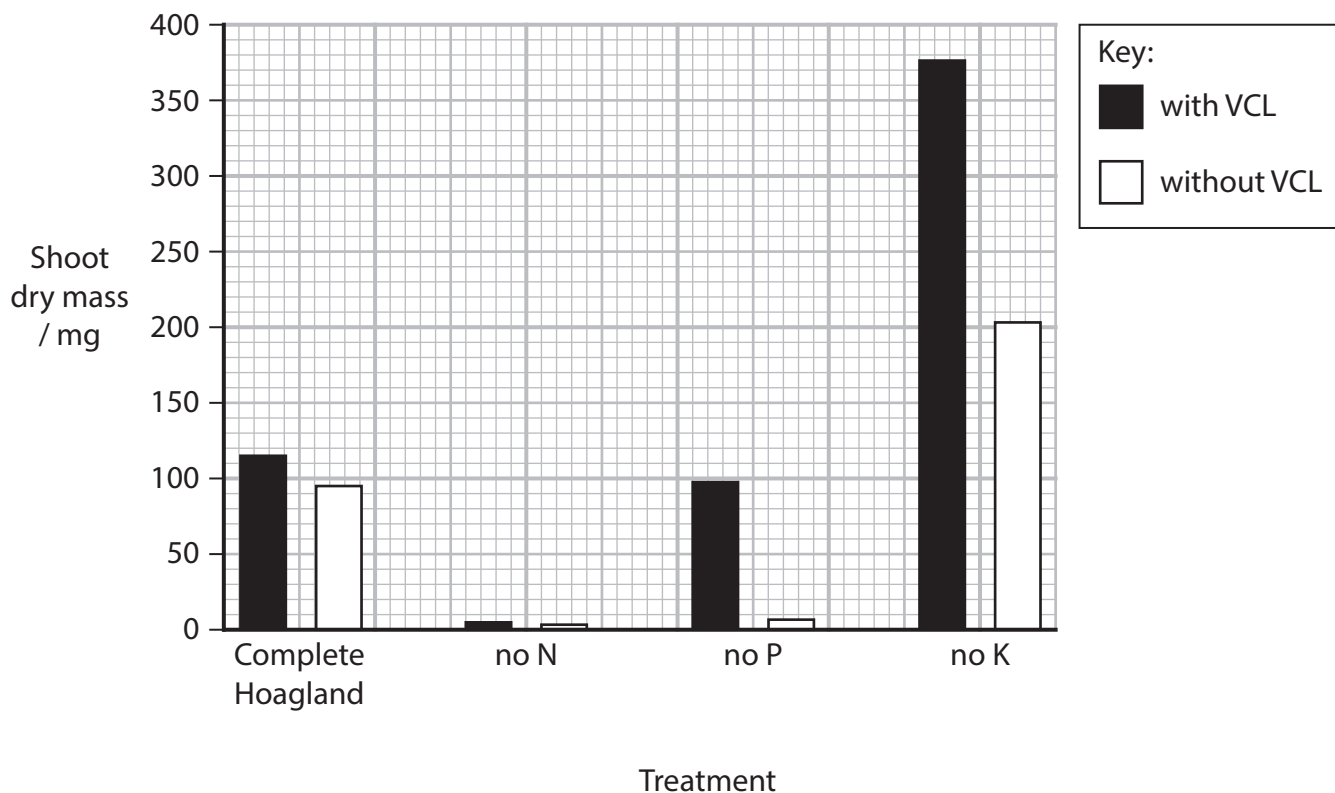


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The graph shows details of the treatments and the results for shoot dry mass.



- (a) (i) Describe a method that could be used to obtain the results for the complete Hoagland treatment, so that it can be compared with the other treatments.

(4)

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- (ii) Calculate the percentage increase in shoot dry mass when VCL is added to Hoagland's solution.

(2)

Answer%

- (b) (i) Suggest why shoot dry mass is a more accurate measure than the length of a fresh shoot.

(2)

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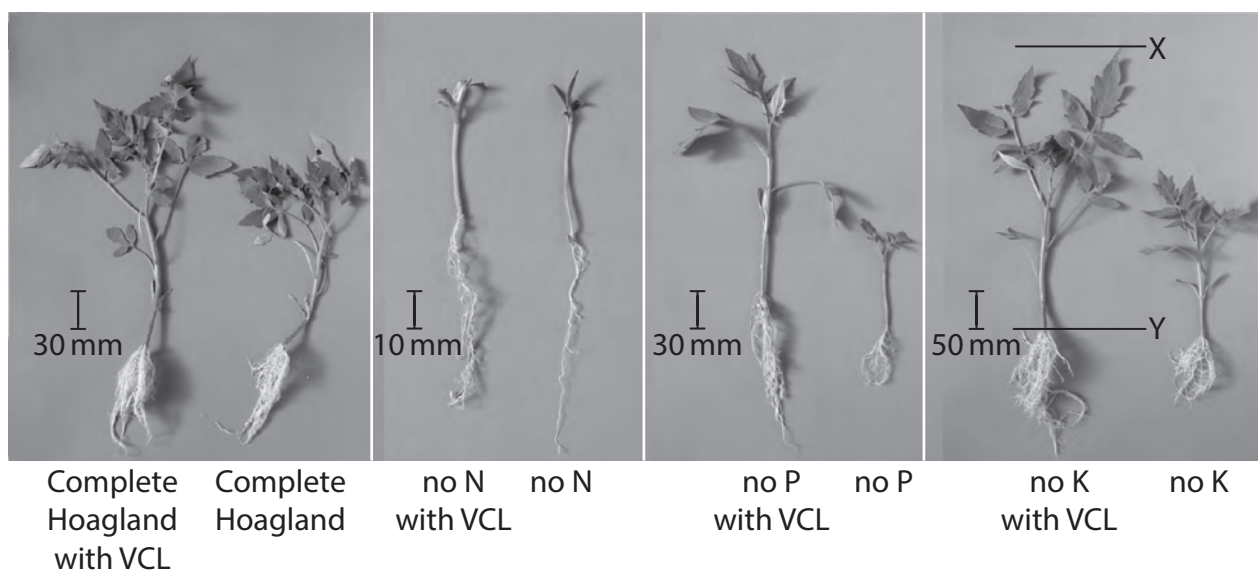
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(ii) The photographs show the appearance of plants following each treatment.

The photograph for each pair of plants has its own scale bar.



Complete the table to show the actual shoot length, between X and Y, for the tomato plant labelled no K, with VCL.

(2)

Actual shoot length / mm							
Complete Hoagland with VCL	Complete Hoagland	no N with VCL	no N	no P with VCL	no P	no K with VCL	no K
240	180	40	40	210	96		230

(iii) Discuss the use of VCL as a fertiliser.

Use all the information from this investigation.

(5)

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(Total for Question 2 = 15 marks)



Source acknowledgment:

Question 2: <https://journals.ashs.org/hortsci/view/journals/hortsci/47/9/article-p1304.xml>
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